

SECTION C

10 (short) A coaxial cable consists of an inner conductor and an outer conductor with radii of $R_1 = 0.8$ mm and $R_2 = 1$ mm, respectively, shown in Fig. 8. The inner conductor is a hollow cylinder of length $L = 2$ m. The gap in between is filled with a dielectric material of relative permittivity $\epsilon_r = 1000$. The inner conductor is biased at $V = 5$ V, while the outer conductor is connected to earth.

- (a) Calculate the charge on the inner conductor. [6]
- (b) What is the capacitance of the coaxial cable? [4]

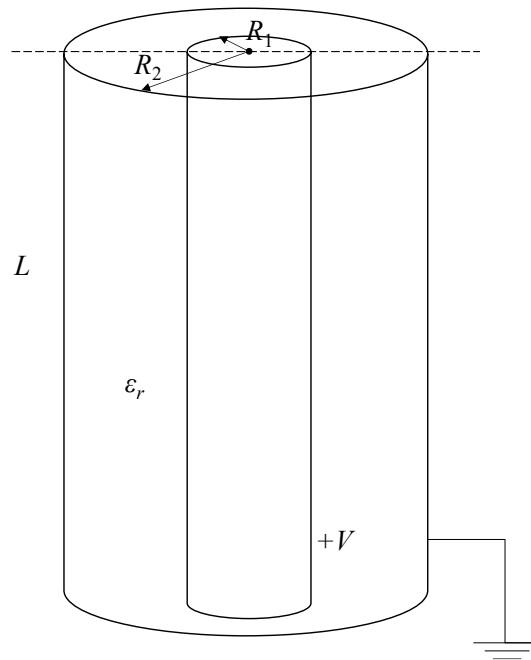


Fig. 8